CIRCLE-VALUED MORSE THEORY AND FROBENIUS MONODROMY

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Abstract. Given a closed manifold M and a circle-valued Morse function $f : M \to S^1$, Hutchings and Lee define a topological invariant $\overline{I}(f) = \overline{\tau}(f) \cdot \overline{\zeta}(f)$ of the homotopy class of f by incorporating data of critical points of f along with orbits. Motivated by Treumann's work on local systems on S^1 with Frobenius monodromy, we construct a characteristic p analogue $\overline{I}_k(f)$ valued in the Whitehead group $K_1(\hat{R})$, where $\hat{R}$ is the Frobenius-skew ring of Laurent series over a perfect field k. We compute $K_1(\hat{R})$ for finite fields and relate our invariant to the Hutchings-Lee invariant, concluding with applications in orbit period-detection.

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1. Introduction

Let M be a smooth finite-dimensional compact manifold. In [HL99], [Hut02], Hutchings and Lee develop an topological invariant associated to any first cohomology class of our manifold. More specifically, the invariant built only depends on the cohomology class $\alpha \in H^1(M;\mathbb{Z})$ and a choice of covering space $\pi : \tilde{M} \to M$ for which $\pi^*\alpha = 0$.

In our setup, consider a circle-valued Morse function $f : M \to S^1$. The Hutchings-Lee invariant specializes to an invariant only depending on the homotopy class of f by picking $\tilde{M}$ to be a connected component of the covering space $M \times_{S^1} \mathbb{R}$ where $\rho : \mathbb{R} \to S^1$ is the usual universal cover $t \mapsto \exp(2\pi i t)$.

In what follows, the (classical) Novikov rings we will consider are all subrings of $\mathbb{Z}((t))$ of the form $\mathbb{Z}((t^n))$ for some n, where n is the degree, in the target S^1, of the generator h of the Deck group of our cover.

The invariant, which we denote by $\overline{I}(f)$ or $\overline{I}$ if the context is clear, is defined as:

$$\overline{I}(f) := \tau(f) \cdot \zeta(f) \in \mathbb{Q}((t))/(\pm t^n)$$

where we are formally quotienting by a group action of $\mathbb{Z} \oplus \mathbb{Z}/2$ given by left multiplication by $\pm t$. If K is a fibered knot, the invariant associated to the fibration on the 0-surgery of the knot complement of $K \subset S^3$ recovers the Alexander polynomial

$$\overline{I} := \Delta_K(t)/(1-t)^2$$

This is closely related to previously known results from Turaev.

On the other hand, Treumann shows in [Tre17] uses the Alexander polynomial to count isomorphism classes of a locally constant sheaf on S^1 with stalks given by a perfect field k of characteristic $p > 0$ and monodromy given by the Frobenius isomorphism $\sigma : x \mapsto x^p$.

In this paper, we take inspiration from Treumann's approach by building an analogue of $\overline{I}$ in a characteristic p ring over a perfect field k twisted by the Frobenius automorphism $\sigma : k \to k$ sending $x \mapsto x^p$. We show how this invariant is related to the classical Hutchings-Lee invariant, and use it for orbit detection.

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2. Setup

Let R_k be the noncommutative ring of σ-skew-Laurent polynomials over k, which we formally define at the set of almost-null $\mathbb{Z}$-indexed sequence of elements in k with pointwise addition and multiplication given by:

$$((x_i)_{i\in\mathbb{Z}} \cdot (y_j)_{j\in\mathbb{Z}})_l := \sum_{i+j=l} x_i\sigma^i(y_j) \in k$$

We denote the indeterminate $t = (\delta_{i1})_{i\in\mathbb{Z}} \in R_k$, so that any sequence $(a_i)_{i\in\mathbb{Z}} \in R_k$ of all but finitely many nonzero elements can be written as:

$$\sum_{i\in\mathbb{Z}} a_i t^i \in R_k$$

(Mind the convention that t is on the right). The ring R can be described by $k[t,t^{-1}]/\{x \in k \ ; \ tx = \sigma(x)t\}$. The idea is that a multiplication by t is akin to applying the monodromy σ at the stalk k. Notice furthermore the injection:

$$\mathbb{F}_p[t,t^{-1}] = k^\sigma[t,t^{-1}] \hookrightarrow R_k$$

Let $R_{k,+}$ be the subring of R_k with $a_i = 0$ for $i < 0$. Denote by $\hat{R}_{k,+}$ the completion of $R_{k,+}$ with respect to the two-sided ideal (t), and let $\hat{R}_k$ be the localization of $\hat{R}_{k,+}$ at t, giving the ring of σ-skew-Laurent series over k. Here $\hat{R}_k$ denotes a Frobenius-skew analogue of the Novikov ring, which is classically given by $\mathbb{Z}((t))$ in the case described above. Notice that $\hat{R}_k$ is a division algebra, so in particular Dieudonné determinants explicitly give an isomorphism from the first K-group of $\hat{R}_k$ to $\hat{R}_k^\times/[\hat{R}_k^\times,\hat{R}_k^\times]$, see Corollary 2.2.6 in [Ros94]. If the context is clear, we will drop the k subscript in the notation of $\hat{R}_k$ (resp. $R_k, R_{k,+}, \hat{R}_{k,+}$).

We then have ring maps:

$$\mathbb{Z}((t)) \to \mathbb{F}_p((t)) \to \hat{R}$$

which realizes $\hat{R}$ as a $\mathbb{F}_p((t))$-module, with the action given by multiplication on the *right*. Since $\mathbb{F}_p((t))$ is a field, $\hat{R}$ is flat over $\mathbb{F}_p((t))$.

Let $\tilde{\mathcal{B}} \subset \tilde{M}$ be a fixed choice of lifts of critical points of M and let $\tilde{\mathcal{B}}_i$ denote the critical points of index i. Fix a metric g such that the pair (df, g) is admissible (in the sense of [Hut02]). Denote by h the generator of the infinite cyclic group of Deck transformations $\mathrm{Aut}_M(\tilde{M}) \cong \mathbb{Z}h$ with the sign of h chosen so that $\tilde{f}(\tilde{p}) - \tilde{f}(h\tilde{p}) > 0$ for $\tilde{p} \in M$ and $\tilde{f} : \tilde{M} \to \mathbb{R}$ the lift of f. In what follows, let $d > 0$ denote the degree of h as a loop in the target S^1 in absolute value: It corresponds to how many times

the generator of the cyclic cover wraps around the target S^1.

In the objects we define, we denote $\alpha = (df, g)$ the tuple associated to the function $f : M \to S^1$ along with the choice of metric g. The requirements in [Hut02] is that $\alpha = (df, g)$ is admissible (Morse-Smale with no degenerate orbits).

Definition 2.1. We define the *Frobenius twisted Novikov complex* as the chain complex of finite free left $\hat{R}$-modules:

$$CN_i(M, \alpha; \hat{R}) := \bigoplus_{\tilde{p} \in \mathcal{B}_i} \hat{R}\tilde{p}$$

with differential $\partial_i : CN_i(M, \alpha; \hat{R}) \to CN_{i-1}(M, \alpha; \hat{R})$ defined on generators by:

$$\partial_i \tilde{p} := \sum_{\tilde{q} \in \mathcal{B}_{i-1}} \sum_{k \geqslant 0} N_X(\tilde{p}, h^k \tilde{q}) \cdot t^{dk} \tilde{q}$$

where $N_X(\tilde{p}, h^k\tilde{q})$ denotes the signed count of flow lines from $\tilde{p}$ to $h^k\tilde{q}$. We abusively denote ∂_i by ∂.

Lemma 2.2. *The twisted Novikov complex is a chain complex.*

Proof. First we do this for $d = 1$. Since ∂ is extended (left) linearly by the $\hat{R}$-module, it suffices to check that $\partial^2\tilde{p} = 0$. But then:

$$\begin{aligned}
\partial^2\tilde{p} &= \sum_{\tilde{q} \in \mathcal{B}_{i-1}} \sum_{k \geqslant 0} N_X(\tilde{p}, h^k\tilde{q}) t^k \sum_{\tilde{r} \in \mathcal{B}_{i-2}} \sum_{j \geqslant 0} N_X(\tilde{q}, h^j\tilde{r}) t^j \tilde{r} \\
&= \sum_{\tilde{r} \in \mathcal{B}_{i-2}} \sum_{l \geqslant 0} \left(\sum_{\tilde{q} \in \mathcal{B}_{i-1}} \sum_{k=0}^{l} N_X(\tilde{p}, h^k\tilde{q}) N_X(h^k\tilde{q}, h^l\tilde{r}) \right) \cdot t^l \tilde{r}
\end{aligned}$$

since flow equivariance gives $N_X(\tilde{q}, h^j\tilde{r}) = N_x(h^k\tilde{q}, h^{j+k}\tilde{r})$ and since each count $N_X(-,-) \in \mathbb{F}_p = k^\sigma$. Now the term in the parenthesis is precisely the signed count of the boundary of the 1-dimensional manifold of flow lines from $\tilde{p}$ to $h^l\tilde{r}$, which is hence 0. For higher d's the same argument applies since $\mathbb{F}_p$ is also fixed by σ^d. □

We provide another interpretation of the twisted Novikov complex. If $CN_*(M, \alpha)$ is the classical Novikov complex, then one can define:

$$CN_*(M, \alpha; \mathbb{F}_p) := CN_*(M, \alpha) \otimes_{\mathbb{Z}((t))} \mathbb{F}_p((t))$$

as a complex of free finite $\mathbb{F}_p((t))$-modules. We then have, by the definition of the right action on the twisted Novikov complex:

$$CN_*(M, \alpha; \hat{R}) \cong \hat{R} \otimes_{\mathbb{F}_p((t))} CN_*(M, \alpha; \mathbb{F}_p)$$

as a complex of abelian groups. In particular, acyclicity of the twisted Novikov complex reduces to acyclicity of $CN_*(M, \alpha; \mathbb{F}_p)$.

In [Hut02], torsion is defined (for the specific case of the infinite cyclic cover) by passing to the fraction field of $\mathbb{Z}((t))$ which is a subfield of $\mathbb{Q}((t))$:

$$CN_*(M, \alpha; \mathbb{Q}) := CN_*(M, \alpha) \otimes_{\mathbb{Z}((t))} \mathbb{Q}((t))$$

The general computation of torsion from the Novikov complex is done by passing to field summands of the fraction ring of the Novikov ring. In particular, in cases of interest, $CN_*(M,\alpha)$ might not be acyclic even if $CN_*(M,\alpha;\mathbb{Q})$ is.

Lemma 2.3. *If $CN_*(M,\alpha;\mathbb{Q})$ is acyclic, then there exists finitely many primes $p_1,\cdots,p_n$ for which*

$$CN_*(M,\alpha;\mathbb{Z}_P) := CN_*(M,\alpha)\otimes_{\mathbb{Z}((t))}\mathbb{Z}_P((t))$$

is acyclic for $P = p_1\cdots p_n$. In particular, this implies that $CN_*(M,\alpha;\mathbb{F}_p)$ is acyclic for all but finitely many primes p.

Before proving this, we remark that $\mathbb{Z}[\![t]\!]$ is a UFD ([Sam],2) of dimension 2 where all maximal ideals are of the form (p,t) for p prime (If a maximal ideal does not contain t, then adding t gives a unit, i.e a term with constant coefficient ± 1, but adding t does not affect the constant coefficient, so the unit must also live in the original maximal ideal). So $\mathbb{Z}((t))$ is a 1-dimensional UFD and hence a PID.

Proof. Each $M_k := H_k(CN_*(M,\alpha))$ is a finitely generated module over the PID $\mathbb{Z}((t))$, hence inducing an isomorphism by the structure theorem for finitely generated modules over a PID:

$$M_k \cong \mathbb{Z}((t))^{r_k}\oplus\frac{\mathbb{Z}((t))}{(p_{k,1}(t))}\oplus\cdots\oplus\frac{\mathbb{Z}((t))}{(p_{k,i_k}(t))}$$

Now since $\mathbb{Q}((t))$ is flat over $\mathbb{Z}((t))$ then:

$$H_k(CN_*(M,\alpha;\mathbb{Q}))\cong M_k\otimes_{\mathbb{Z}((t))}\mathbb{Q}((t))$$

so acyclicity of the former forces precisely $r_k = 0$ for all k. So let P denote the product of all prime factors of leading coefficients of $\{p_{i,j}(t)\}_{0\leqslant i\leqslant n,1\leqslant j\leqslant i_k}$ (where $n = \dim M$). In that case we see that each $p_{i,j}(t)$ becomes a unit in $\mathbb{Z}_P((t))$ and hence since $\mathbb{Z}_P((t))$ is flat over $\mathbb{Z}((t))$ (since it's torsion free) the same argument shows $CN_*(M,\alpha;\mathbb{Z}_P)$ is acyclic, as desired.

The last conclusion comes by taking the short exact sequence:

$$0\to CN_*(M,\alpha;\mathbb{Z}_P)\xrightarrow{\times p} CN_*(M,\alpha;\mathbb{Z}_P)\to CN_*(M,\alpha;\mathbb{F}_p)\to 0$$

and taking long exact sequence in homology. □

Remark 2.4. If one only cares about homology mod p, the above argument can be refined as follows: Since $\mathbb{Z}((t))$ is a PID, UCT applies ([Wei94] 3.6.2) to give a short exact sequence:

$$0\to M_k\otimes_{\mathbb{Z}((t))}\mathbb{F}_p((t))\to H_k(CN_*(M,\alpha;\mathbb{F}_p))\to \mathrm{Tor}_1^{\mathbb{Z}((t))}(M_{k-1},\mathbb{F}_p((t)))\to 0$$

Additivity of Tor and [Eis95], A3.17 yield:

$$\mathrm{Tor}_1^{\mathbb{Z}((t))}(M_{k-1},\mathbb{F}_p((t)) = \bigoplus_{j=1}^{i_{k-1}}\frac{(p)\cap(p_{k-1,j}(t))}{(p\cdot p_{k-1,j}(t))}$$

giving us a precise homology vanishing statement: Under the above assumption, $CN_*(M,\alpha;\mathbb{F}_p)$ fails to be acyclic only at primes p which divide all coefficients of some $p_{i,j}(t)$, i.e if and only if there exists a torsion p element in M_k for some k. In particular since $\mathbb{Z}((t))$ is flat over $\mathbb{Z}[t^{\pm 1}]$, we have the following:

$$CN_*(M,\alpha;\mathbb{F}_p) \text{ is not acyclic} \iff \dim_{\mathbb{F}_p} \operatorname{Tor}_1^{\mathbb{Z}}(H_*(\tilde{M}),\mathbb{F}_p) = \infty$$

as $M_k \cong H_k(C_*(\tilde{M}) \otimes_{\mathbb{Z}[t^{\pm}]} \mathbb{Z}((t)))$ (see [Hut02]). Thus a sufficient condition for $CN_*(M,\alpha;\mathbb{F}_p)$ to be acyclic is $\dim_{\mathbb{F}_p} H_*(\tilde{M};\mathbb{F}_p) < \infty$.

So the twisted Novikov complex is nearly always acyclic if the classical one is. Although not a complex of modules over a commutative ring, one can still define Reidemeister torsion over the Frobenius twisted Novikov complex using Dieudonné determinants, see [Kit10].

Definition 2.5. We denote the Reidemeister torsion of $CN_*(M,\alpha;\hat{R})$ (with the choice of basis $\mathcal{B}$) by

$$\tau_k(\alpha,\mathcal{B}) \in K_1(\hat{R}) = \hat{R}^\times/[\hat{R}^\times,\hat{R}^\times]$$

We can also define an analogue of the Zeta function as follows. We cannot use the exponential notation since $\hat{R}$ has characteristic p, so we instead use the product notation. Let $\mathscr{E}$ denote the set of embedded orbits on M of the admissible pair $\alpha = (df, g)$ chosen above. For each such embedded orbit $\gamma \in \mathscr{E}$, we define:

- $w(\gamma) \in \mathbb{N}^*$: The unique positive integer for which the lift of γ at p in $\tilde{M}$ ends in $h^{\pm w(\gamma)}p$ (assuming γ is oriented to go downwards, then the lift would end at $h^{w(\gamma)}p$).
- $a(\gamma)$: The number of eigenvalues > 1 of γ's first return map's differential.
- $b(\gamma)$: The number of eigenvalues < -1 of γ's first return map's differential.

Definition 2.6. In the above setup, define the Zeta function by:

$$\zeta_k(\alpha) := \prod_{\gamma\in\mathscr{E}} (1-(-1)^{b(\gamma)}t^{dw(\gamma)})^{-(-1)^{a(\gamma)+b(\gamma)}} \in \hat{R}^\times$$

For an admissible pair $\alpha = (df, g)$, the above quantity is well-defined since there are finitely many bounded-energy orbits. Another easy way to see that this is well-defined is because there is a natural map $\mathbb{Z}((t))^\times \to \hat{R}^\times$ and, by the above definition, this map sends the classical Hutchings-Lee zeta function $\zeta(\alpha) \in \mathbb{Z}((t))^\times$ to $\zeta_k(\alpha)$ defined above, since the same product formula holds in the classical case.

Using again that each $N_X(-,-) \in \mathbb{F}_p = k^\sigma \subset k$, we see that a change of basis $\mathcal{B}$ only changes $\tau_k(\alpha,-)$ by $\pm t^n$. Thus, we define our invariant to be:

$$\overline{I}_k(\alpha) := \overline{\tau_k(\alpha,\mathcal{B})} \cdot \overline{\zeta_k(\alpha)} \in K_1(\hat{R})/(\pm t^n)$$

where we are again quotienting by multiplication by $\pm t^n$.

Theorem 2.7. *The quantity $\overline{I}_k(\alpha) \in K_1(\hat{R})/(\pm t^n)$ is a topological invariant of M only depending on the homotopy class of $f : M \to S^1$.*

We will provide an à posteriori proof of Theorem 2.7 by making use of invariance as proved in [Hut02]. However, it is possible to do an à priori proof by modifying the proof in [Hut02], with the only notable modification being that one needs to remove usages of logs and exponentials.

3. Proof of invariance

Before proving invariance, we first relate the torsion to the classical Hutchings-Lee torsion in the case where $CN_*(M, \alpha; \mathbb{F}_p)$ is acyclic.

Proposition 3.1. *Let $f : M \to S^1$ be a degree $d > 0$ map (in absolute value) and g be a metric such that $\alpha = (df, g)$ is admissible. If $\overline{\tau}(\alpha) \in \mathbb{Q}((t^d))^\times/(\pm t^n)$ denotes the Reidemeister torsion, then one can find coprime polynomials $G_0, G_1 \in \mathbb{Z}[t]$ $\overline{\tau}(\alpha) = \dfrac{G_0(t)}{G_1(t)}$ and:*

$$\overline{\tau}_{\mathbb{F}_p}(\alpha) = \frac{G_0(t) \pmod p}{G_1(t) \pmod p} \in \mathbb{F}_p((t^d))^\times/(\pm t^n)$$

for every prime p with $CN_*(M, \alpha; \mathbb{F}_p)$ acyclic.

Proof. The fact the Hutchings-Lee invariant can be written as a rational function follows since, for generic choice of metric g in the C^0-topology, the coefficients of the Novikov complex are rational functions (See [Paj06], 12). In what follows, we assume $d = 1$ for simplicity.

Let p be a prime for which $CN_*(M, \alpha; \mathbb{F}_p)$ is acyclic. By the same argument as in Lemma 2.3, we see that acyclicity of $CN_*(M, \alpha; \mathbb{F}_p)$ implies acyclicity of:

$$CN_*(M, \alpha; \mathscr{O}) := CN_*(M, \alpha) \otimes_{\mathbb{Z}((t))} \mathscr{O}$$

where $\mathscr{O}$ is the discrete valuation ring $\mathscr{O} = \mathbb{Z}((t))_{(p)}$. We then have two maps $\mathscr{O}$:

$$\mathbb{Q}((t)) \xleftarrow{\iota} \mathscr{O} \xrightarrow{\pi} \mathbb{F}_p((t))$$

where π is the quotient map and ι factors through the localization map at the generic point of $\mathscr{O}$ which is an inclusion into $\mathrm{Frac}(\mathbb{Z}((t)) \subset \mathbb{Q}((t))$. If $u \in \mathscr{O}^\times$ is the Reidemeister torsion of $CN_*(M, \alpha; \mathscr{O})$ with respect to $\mathcal{B}$, then functoriality implies $\tau(\alpha, \mathcal{B}) = \iota_* u = u \in \mathbb{Q}((t))^\times$ and $\overline{\tau}_{\mathbb{F}_p}(\alpha) = \overline{u} \in \mathbb{F}_p((t))^\times$. In particular, since $u \in \mathscr{O}^\times$ then it has 0 valuation, so p divides neither G_0 nor G_1 (as they are primitive). As desired. □

We now prove Theorem 2.7.

Proof. Acyclicity of $CN_*(M,\alpha;\mathbb{F}_p)$ (and hence $CN_*(M,\alpha;\hat{R})$) only depends on $H_*(\tilde{M})$ and hence only on the homotopy type of the circle-valued Morse function. Invariance of the Hutchings-Lee invariant in $\mathbb{Q}((t))$ descends to $\mathscr{O}^\times = \mathbb{Z}((t))^\times_{(p)}$ and hence to $\mathbb{F}_p((t))^\times/(\pm t^n)$ into $\hat{R}^\times$ via the induced map $\pi_* : \mathscr{O}^\times \to K_1(\hat{R})/(\pm t^n)$. □

As a direct consequence of Proposition 3.1, we see that the invariant $\overline{I}_{\mathbb{F}_p}(f)$ over $\mathbb{F}_p$ can be computed by taking the classical $\overline{I}(f)$ and reducing it mod p in the appropriate sense defined above, so long as $CN_*(M,\alpha;\mathbb{F}_p)$ is acyclic. The key idea for why this holds is that the Hutchings-Lee torsion, for good primes, can be interpreted in terms of the Novikov complex with coefficients in $\mathbb{Z}((t))_{(p)}$.

We now denote $\overline{I}_k(\alpha)$ by $\overline{I}_k(f)$ since the former only depends on the cohomology class $df \in H^1(M;\mathbb{R})$ by the above theorem.

4. Computation of $K_1(\hat{R})$

We compute the Whitehead group:

$$K_1(\hat{R}) = \hat{R}^\times/[\hat{R}^\times,\hat{R}^\times]$$

for specific choices of underlying fields k.

Proposition 4.1. *If $k \cong \mathbb{F}_{p^m}$ for some integer $m \geqslant 1$, then $K_1(\hat{R}) \cong \mathbb{F}_p((t^m))^\times$.*

Proof. Suppose $k \cong \mathbb{F}_{p^m}$ is finite. Then $\hat{R}$ is an index m division algebra over its center $F = Z(\hat{R}) = \mathbb{F}_p((t^m))$, and $E = \mathbb{F}_{p^m}((t^m))$ is a degree m (commutative) field extension of F. Then, $\hat{R}$ is a finite-dimensional central division algebra over the local field F so that the reduced norm map

$$\mathrm{Nrd}_{\hat{R}/F} : \hat{R}^\times \to F^\times$$

has target $F^\times$ and has kernel given by the commutator subgroup $[\hat{R}^\times,\hat{R}^\times]$ as per section 1.4.2-1.4.3 of [PR94]. □

Remark 4.2. The norm map above can be described as follows in the finite-dimensional case. We have that $\hat{R}$ is an m-dimensional vector space over E with basis $1, t, \cdots, t^{m-1}$. The norm map $\mathrm{Nrd}_{\hat{R}/F}$ is then defined by first picking an isomorphism $\hat{R}\otimes_F E \to M_m(E)$ and then taking the determinant of the image $x\otimes 1$ for $x \in \hat{R}$. This procedure does not depend on the chosen isomorphism (See [PR94] section 1.4 for more details).

In our case, the isomorphism $\hat{R}\otimes_F E \to M_m(E)$ can be described as follows: We define for each free tensor $x\otimes e$ a linear endomorphism of the m-dimensional vector space:

$$\hat{R} = \bigoplus_{i=0}^{m-1} Et^i$$

acting by $t^i \mapsto xt^ie$ expressed in said basis. In what follows, for any finite field $k \cong \mathbb{F}_{p^m}$, we let:

$$\mathrm{Nrd}_k : K_1(\hat{R}) \xrightarrow{\cong} \mathbb{F}_p((t^m))^\times$$

denote the norm map described.

Example 4.3. Consider the identity map $S^1 \to S^1$ where we pick $k = \mathbb{F}_{p^2}$. Then it has exactly one orbit, so the invariant is precisely $(1-t)^{-1} \in \hat{R}^\times$ up to $\pm t^n$. Then notice that on the basis $\{1, t\}$, left multiplication by $1-t$ acts via the matrix:

$$\begin{pmatrix} 1 & -t^2 \\ -1 & 1 \end{pmatrix}$$

whose determinant is $1-t^2$, so the invariant in $\mathbb{F}_p((t^2))^\times$ is given by $(1-t^2)^{-1}$, up to t^{2n}.

We now include a discussion on $K_1(\hat{R})$ with underlying field k infinite. First, the group homomorphism $\mathbb{Z} \to \mathrm{Aut}(k^\times)$ sending $1 \mapsto \sigma$ gives us a semidirect product:

$$H = k^\times \rtimes \mathbb{Z}$$

with group law given by $(a,n)\cdot(b,m) = (a\sigma^n(b), nm)$. So we get a group homomorphism:

$$\Psi : \hat{R}^\times \to H$$

sending $\sum_{i \geqslant n} a_i t^i$ (with $a_n \neq 0$) to (a_n, n). Then the kernel of Ψ is $R_\sharp := 1 + t\hat{R}_+$.

Lemma 4.4. *If k either contains a transcendental element or is equal to $\overline{\mathbb{F}}_p$ then $R_\sharp \subset [\hat{R}^\times, \hat{R}^\times]$.*

Proof. First suppose k has a transcendental element say $a_0 \neq 0$. Let $B = \sum_{i \geqslant 0} b_i t^i$ a to-be-determined power series with $b_0 \neq 0$ so that B is invertible with inverse $B^{-1} = \sum_{j \geqslant 0} c_j t^j$ with c_j satisfying $c_0 = b_0^{-1}$ and $c_j = -b_0^{-1} \sum_{p=1}^k b_p \sigma^p(c_{k-p})$. Then in the commutator:

$$a_0^{-1} \cdot B \cdot a_0 \cdot B^{-1} = \sum_{k \geqslant 0} \left(\sum_{i+j=k} a_0^{-1} \sigma^i(a_0) b_i \sigma^i(c_j) \right) t^k$$

the coefficient at $k = 0$ is precisely 1 and, for $k > 0$, the term b_k first appears in coefficient t^k and has factor:

$$\sigma^k(b_0^{-1}) \left(a_0^{-1} \sigma^k(a_0) - 1 \right) \neq 0$$

since a_0 is transcendental. So one can always choose an appropriate b_k to realize this commutator as any term of $R_\sharp$, hence $R_\sharp = [\hat{R}^\times, \hat{R}^\times]$, as desired.

If $k = \overline{\mathbb{F}_p}$, notice instead that $tBt^{-1}B^{-1} = \sigma(B) \cdot B^{-1}$ (where B is the same power series as above with $b_0 = 1$). But the map $R_\sharp \to R_\sharp$ sending $B \mapsto \sigma(B) \cdot B^{-1}$ is surjective: The b_k term first appears in the t^k-coefficient precisely as:

$$\sigma(b_k) - b_k$$

and the equation $b_k^p - b_k - x = 0$ is solvable in $\overline{\mathbb{F}_p}$ for any $x \in \overline{\mathbb{F}_p}$ by algebraic closure. □

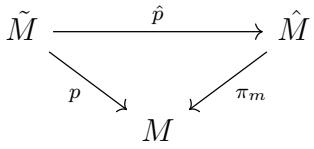
Remark 4.5. In particular notice that if $k = \overline{\mathbb{F}_p}$, the argument does not need $b_0 = 1$ and hence the valuation map yields an isomorphism $K_1(\hat{R}) \cong \mathbb{Z}$.

5. Relating $k = \mathbb{F}_{p^m}$ with m-fold cover

In [Hut02], the norm map serves as the algebraic tool needed to pass to a finite cyclic cover. More precisely, let M be a smooth compact manifold as above and let $f : M \to S^1$ be a circle-valued Morse function and $p : \tilde{M} \to M$ be a connected component of $M \times_{S^1} \mathbb{R}$. Then as remarked above, the Deck transformations are given by $\mathrm{Aut}_M(\tilde{M}) \cong \mathbb{Z}$ (with generator denoted by h) and we have a short exact sequence of abelian groups:

$$0 \to m\mathbb{Z} \to \mathrm{Aut}_M(\tilde{M}) \to \mathbb{Z}/m \to 0$$

Quotienting $\tilde{M}$ by the action of $m\mathbb{Z} \subset \mathrm{Aut}_M(\tilde{M})$ hence gives an m-fold cover $\hat{M}$:



In both cases, the Novikov rings of $\tilde{p}$ and p are integral Laurent series. We have the inclusion of Novikov rings induced from the map $\mathrm{Aut}_{\hat{M}}(\tilde{M}) \cong m\mathbb{Z} \hookrightarrow \mathrm{Aut}_M(\tilde{M})$ given by:

$$\mathbb{Z}((t^{dm})) \hookrightarrow \mathbb{Z}((t^d))$$

realizing $\mathbb{Z}((t^d))$ as a free $\mathbb{Z}((t^{dm}))$-module of rank m (Recall that t is attached to the target S^1). For each $f(t^d) \in \mathbb{Z}((t^d))$, right multiplication by $f(t^d)$ gives a map of rank m free $\mathbb{Z}((t^{dm}))$-modules whose determinant we denote by $\mathrm{Norm}(f(t^d))$. The above construction naturally respects passage to the fraction field, so we can define the same norm map for elements of $\mathbb{Q}((t^d))$.

Having fixed a map $f : M \to S^1$ at the start for which critical points are non-degenerate, we naturally deduce that critical points of the lifted map $\hat{f} : \hat{M} \to S^1$ are also non-degenerate. Let $\alpha = (df, g)$ be admissible for M so that $\hat{\alpha} = (d\hat{f}, \hat{g})$ (where $\hat{g}$ denotes the pullback of g to $\hat{M}$) is also admissible.

Theorem 5.1. *(Hutchings) The invariant $\overline{I}(\hat{f}) \in \mathbb{Q}((t^{dm}))/(\pm t^n)$ of the m-fold cover is the image of $\overline{I}(f) \in \mathbb{Q}((t^d))/(\pm t^n)$ under the norm map.*

Proof. See 3.5-3.6 in [Hut02]. □

Remark 5.2. The above procedure is clearly well-defined since the norm map is multiplicative and $\mathrm{Norm}(t^d) = t^{dm}$, and G acts on $\mathbb{Q}((t^{dm}))$ precisely by multiplication $\pm t^{dm}$.

Example 5.3. Consider a map $f : S^1 \to S^1$ which has two critical points p, q and no orbits. The Novikov complex looks like:

$$0 \to \mathbb{Z}((t)) \xrightarrow{\times(1-t)} \mathbb{Z}((t)) \to 0$$

and has torsion $(1-t)^{-1}$, while the Novikov complex of the double cover is:

$$0 \to \mathbb{Z}((t))^{\oplus 2} \xrightarrow{\begin{pmatrix} 1 & -t \\ -t & 1 \end{pmatrix}} \mathbb{Z}((t))^{\oplus 2} \to 0$$

and has torsion $(1-t^2)^{-1}$. Multiplication by $1-t$ acts on the basis $1, t$ of $\mathbb{Z}((t))$ viewed as a rank 2 free $\mathbb{Z}((t^2))$-module by:

$$\begin{pmatrix} 1 & -t^2 \\ -1 & 1 \end{pmatrix}$$

which has determinant $1-t^2$, as we would expect from Theorem 5.1.

Notice that Theorem 5.1 and the computation of the norm map in Proposition 4.1 are the same for finite fields, which suggests the following:

Theorem 5.4. *Let $f : M \to S^1$ and $\hat{f} : \hat{M} \to S^1$ the corresponding map for the m-fold cover. Let $\alpha = (df, g)$ be an admissible pair and let $\hat{\alpha} = (d\hat{f}, \hat{g})$ be its lift to the m-fold cover. If $CN_*(M, \alpha; \hat{R}_{\mathbb{F}_{p^m}})$ is acyclic, then so is $CN_*(\hat{M}, \hat{\alpha}; \hat{R}_{\mathbb{F}_p})$ and:*

$$\mathrm{Nrd}_{\mathbb{F}_{p^m}}(\overline{I}_{\mathbb{F}_{p^m}}(f)) = \overline{I}_{\mathbb{F}_p}(\hat{f}) \in \mathbb{F}_p((t^m))^\times/(\pm t^n)$$

Proof. Since $CN_*(M, \alpha; \hat{R}_{\mathbb{F}_P^m})$ is acyclic then so is $CN_*(M, \alpha; \mathbb{F}_p)$. As in Proposition 3.6 of [Hut02], we can hence write it as a summand of length 2 acyclic finite free complexes. So the same argument runs for torsion. For the zeta side, we use the multiplicative form, which yields:

$$\overline{I}_{\mathbb{F}_p}(\hat{f}) = \mathrm{Norm}(\overline{I}_{\mathbb{F}_p}(t)) = \mathrm{Norm}(\overline{I}_{\mathbb{F}_{p^m}}(t))$$

where the last equality uses the fact that the invariant is an algebraic function of the coefficients in the Novikov complex and hence lives in the image of $\mathbb{F}_p((t))^\times \to K_1(\hat{R})$ (so that it does not depend on the underlying field). Finally, we see by the computation in Proposition 4.1 for finite fields:

$$\mathrm{Norm}(\overline{I}_{\mathbb{F}_{p^m}}(t)) = \mathrm{Nrd}_{\mathbb{F}_{p^m}}(\overline{I}_{\mathbb{F}_{p^m}}(t))$$

as desired. □

Thus, Theorem 5.4 combined with Proposition 3.1 provides a way of computing our invariant over finite fields.

Example 5.5. The Hutchings-Lee invariant for the figure eight knot is given by:

$$\frac{t^2 - 3t + 1}{(1-t)^2} \in \mathbb{Q}((t))^\times/(\pm t^n)$$

so the invariant reduced mod 2 gives:

$$\frac{t^2+t+1}{(1+t)^2} \in \mathbb{F}_2((t))^\times/(\pm t^n)$$

6. Application to orbit detection

We showcase how mod p information can be used for orbit detection. Continuing on Example 5.5 with K the figure eight knot, suppose that the natural map $S^0(K) \to S^1$ on the 0-surgery is homotopy equivalent to a map $f : S^0(K) \to S^1$ with no critical points. Invariance mod 2 hence gives an equality:

$$\frac{t^2+t+1}{(1+t)^2} = \prod_{\gamma\in\mathscr{E}}(1+t^{w(\gamma)})^{-(-1)^{a(\gamma)+b(\gamma)}} \in \mathbb{F}_2[\![t]\!]$$

for $\mathscr{E}$ the set of embedded orbits of f under any choice of admissible metric. By admissibility, there are finitely many orbits γ with $w(\gamma) \leqslant 4$, we hence get an equality:

$$\frac{t^2+t+1}{(1+t)^2} = \prod_{\gamma\in\mathscr{E}, w(\gamma)\leqslant 3}(1+t^{w(\gamma)})^{-(-1)^{a(\gamma)+b(\gamma)}} \in (\mathbb{F}_2[\![t]\!]/(t^4))^\times$$

Since $(1+t^2) = (1+t)^2$, then notice that if there is no γ with $w(\gamma) = 3$, that would imply t^2+t+1 is in the subgroup of the group of units of $\mathbb{F}_2[\![t]\!]$ generated by $1+t$, which is $\langle 1+t\rangle = \{1, 1+t, 1+t^2, 1+t+t^2+t^3\}$. So there exists γ with $w(\gamma) = 3$.

More generally, we have the following: If $f : M \to S^1$ be a smooth degree 1 map with no critical points, and g is any metric for which (df, g) is admissible (i.e no degenerate orbits), then for any $N \in \mathbb{N}$, reduction gives an equality:

$$\overline{\zeta}_{\mathbb{F}_2}(f) = \prod_{\gamma\in\mathscr{E}\ ;\ w(\gamma)\leqslant N}(1+t^{w(\gamma))})^{-(-1)^{a(\gamma)+b(\gamma)}} \in (\mathbb{F}_2[[t]]/(t^{N+1}))^\times$$

In particular, if $\overline{\zeta}_{\mathbb{F}_2}(f)$ does not lie in the subgroup of units generated by $1+t^i$ for $i \leqslant N-1$ then there exists an orbit of period exactly N. In particular, this criterion is independent of the chosen metric g apart from admissibility.

Example 6.1. Let $M_0 = \#^3(S^3 \times S^3)$ be the threefold connected sum of $S^3 \times S^3$, so that $H_0(M;\mathbb{Z}) \cong H_6(M;\mathbb{Z}) \cong \mathbb{Z}$ and $H_3(M;\mathbb{Z}) \cong \mathbb{Z}^6$. Consider the matrix:

$$A := \begin{pmatrix} -4 & 0 & 0 & -4 & 1 & 0 \\ 1 & 4 & 4 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 \\ 4 & -1 & -1 & 4 & -2 & 0 \\ 0 & -4 & -4 & 1 & -4 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}$$

By a theorem of [Kra20], this matrix can be realized as the induced map on H_3 by some diffeomorphism ϕ of M_0. Letting M be the mapping torus of ϕ, we see that the Hutchings-Lee invariant associated to the projection map $M \to S^1$ is given by

Lefschetz' Theorem by $\det(A - tI) \cdot (1-t)^{-2} = (t^6 - t^4 - t^3 - t^2 + 1) \cdot (1-t)^{-2}$. In particular, mod 2 we get:

$$\frac{(t^2+t+1)(t^4+t^3+t^2+t+1)}{(1+t)^2} = \frac{(1+t^3)(1+t^5)}{(1+t^4)} \in \mathbb{F}_2((t))^\times$$

Reducing modulo t^4, we see that the subgroup of units in $\mathbb{F}_2[\![t]\!]$. We hence get an equality:

$$\frac{(1+t^3)(1+t^5)}{(1+t^4)} = \prod_{\gamma \in \mathscr{O}} (1+t^{w(\gamma)})^{-a(\gamma)+b(\gamma)} \in \mathbb{F}_2((t))^\times$$

Reducing mod t^4, the subgroup of units in $\mathbb{F}_2[\![t]\!]/(t^4)$ generated by $1+t, 1+t^2$ is $\{1, 1+t, 1+t^2, 1+t+t^2+t^3\}$ forcing the existence of γ with $w(\gamma) = 3$ (i.e a period 3 orbit).

Similarly, reducing mod t^6, let H be the subgroup of units in $\mathbb{F}_2[\![t]\!]/(t^6)$ generated by $\langle 1+t, 1+t^3 \rangle$ (since $1+t^2 = (1+t)^2$ and $1+t^4 = (1+t)^4$). If there were no period 5 orbits, then we would get $1+t^5 \in H$, which is not the case by a casework.

Thus, we deduce that ϕ (generically) admits a fixed point of order exactly 3 and a fixed point of order exactly 5.

Remark 6.2. The matrix A above was found computationally by seeking low trace values for $\mathrm{tr}(A), \mathrm{tr}(A^3), \mathrm{tr}(A^5)$. In the above example, we have $\mathrm{tr}(A) = 0, \mathrm{tr}(A^3) = 3$ and $\mathrm{tr}(A^5) = 5$, so the naive Lefschetz count gives $\#\mathrm{Fix}(\phi) = 2$, $\#\mathrm{Fix}(\phi^3) = -1$ and hence does not detect if the fixed point of ϕ^3 is period 1 or 3.

Further directions. One natural question from our analysis is whether meaningful topological information can be extracted from bad primes p for which the complex $CN_*(M, f; \mathbb{F}_p)$ is not acyclic.

Another possible direction would be to try to incorporate homological information of fibers $f^{-1}(t)$ at regular values t. For example, the Hutchings-Lee invariant on a perturbation of the projection map $S^2 \times S^1 \to S^1$ is $1/(1-t)^2$ while the zeta function of $\mathbb{P}^1_{\mathbb{F}_p}$ is $1/(1-t)(1-pt)$, so an analogue would require distinguishing the two fixed points of the flow map on the fiber S^2.

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